# H2 Math Summary

Notes

Maximus Pung

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# 1 Systems of Linear Equations

## 1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

**Example 1.1**

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

**Example 1.2**

The "variables" may be expressions. To find $x$, $y$ and $z$, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && —(1)\\ x^3-\ln y+2\mathrm{e}^z&=9 && —(2)\\ x^3+2\ln y+\mathrm{e}^z&=11 && —(3)\end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in $A$, $B$ and $C$. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because $x$ appears as both $x^2$ and $x$, and there are product terms such as $xz$ and $xy$.

## 1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

**Example 1.3**

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for $\lambda$ and $\mu$ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

## 1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

**Example 1.4**

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and $z$ can be any real value. So just sub that into $x$ and $y$ to get a correct solution.

$z=-1$: $(7,0,-1)$ $z=0$: $(\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1$: $(4,1,1)$ $z=2$: $(\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3$: $(1,2,3)$ $z=4$: $(-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5$: $(-2,3,5)$ $z=6$: $(-\tfrac{7}{2},\tfrac{7}{2},6)$

We can use the GC table to find all these solutions.

# 2 Inequalities

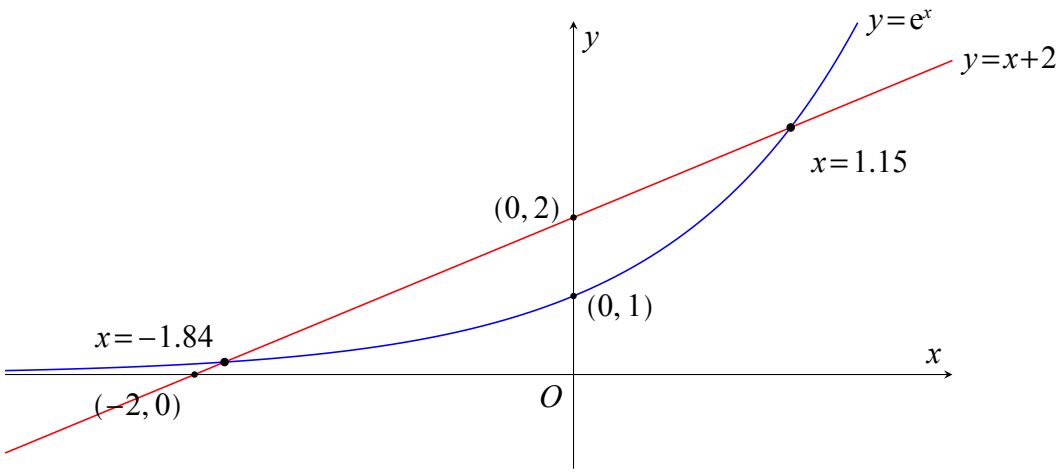
## 2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the $x$-values where the required graph is above or below.

**Example 2.1**

$$e^x < x + 2$$




Using GC, intersections at $x = -1.8414$, $x = 1.1462$

$$\therefore\ -1.84 < x < 1.15$$

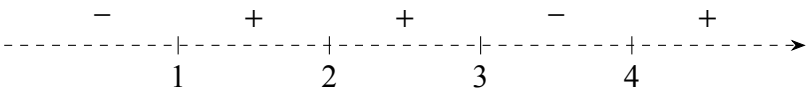
## 2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant $<0$ and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

**Example 2.2**

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$




$x=2$ : numerator $=0$ (included)

$x=3,\ x=4$ : denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

You can sketch the graph to check the answer.

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the sign the same.

## 2.3 Common Substitutions

Once an inequality in $x$ is solved, we can replace $x$ with an expression to deduce the solution of a related inequality.

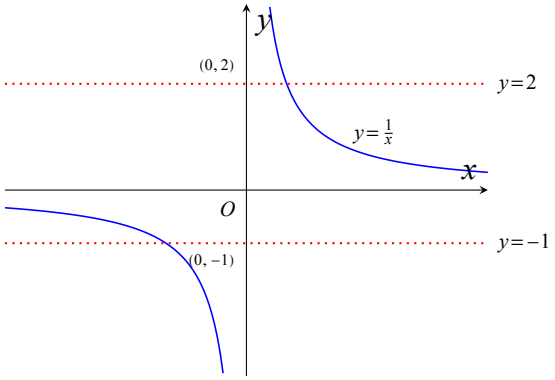
| **Replace $x$ by** | **Original inequality** | | |
|---|---|---|---|
| | $x>2$ | $x<-1$ | $-1<x<2$ |
| $e^x$ | $x>\ln 2$ | no solution | $x<\ln 2$ |
| $\ln x$ | $x>e^2$ | $0<x<e^{-1}$ | $e^{-1}<x<e^2$ |
| $-x$ | $x<-2$ | $x>1$ | $-2<x<1$ |
| $x+1$ | $x>1$ | $x<-2$ | $-2<x<1$ |
| $\lvert x\rvert$ | $x<-2$ or $x>2$ | no solution | $-2<x<2$ |
| $x^2$ | $x<-\sqrt{2}$ or $x>\sqrt{2}$ | no solution | $-\sqrt{2}<x<\sqrt{2}$ |
| $\frac{1}{x}$ | $\left(0<x<\frac{1}{2}\right)$ | $(-1<x<0)$ | $\left(x<-1 \text{ or } x>\frac{1}{2}\right)$ |
| $\cos x,\ 0\leqslant x\leqslant 2\pi$ | (no solution) | (no solution) | $(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$ |

Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

**Example 2.3**

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$




**Example 2.4**

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

